

Dmitrii Vasilenko

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RESEARCH INTERESTS

Theory of deep learning, attention and Transformer dynamics, Bayesian deep learning.

EDUCATION

Moscow Institute of Physics and Technology (MIPT)	Moscow, Russia
<i>M.Sc. in Computer Science (GPA: 4.9/5.0)</i>	Sep. 2025 – Jun. 2027 (expected)
Yandex School of Data Analysis (YSDA)	Moscow, Russia
<i>Two-year graduate program in Data Science, admission rate $\approx 8\%$ (GPA: 5.0/5.0)</i>	Sep. 2025 – Jun. 2027 (expected)
Moscow Institute of Physics and Technology (MIPT)	Moscow, Russia
<i>B.Sc. in Technical Physics (GPA: 4.5/5.0)</i>	Sep. 2021 – July 2025

PUBLICATIONS

- Sink vs. Diagonal Attention: Sharpened Cost Bounds and Comparison Regimes** Sep. 2026
– D. Vasilenko, A. Grabovoy. Accepted at the AXIOM Workshop (Foundations of Efficient Deep Learning) @ NeurIPS 2026, Paris. OpenReview.
– Proved an $\Omega(T)$ lower bound on the MLP cost of sink attention under the zero-BOS-value assumption, sharpened the diagonal lower bound by $\sqrt{2}$, and characterized the sink cost as $\Theta(\kappa\sqrt{T} + T)$ at $d = T + 4$.
- Jacobian Analysis of a Recurrent Transformer Block** | AINL 2026 Apr. 2026
– D. Vasilenko, I. Stepanov, V. Kasiuk, G. Karpeev, A. Grabovoy. Presented as a contributed talk at AINL 2026.
– Derived explicit upper bounds on the spectral norm of the full-step Jacobian of Recurrent Transformers; proved and empirically validated that RMSNorm acts as a multiplicative damping factor keeping the Jacobian norm $\mathcal{O}(1)$.

PREPRINTS AND MANUSCRIPTS UNDER REVIEW

- The Parameter Cost of Attention Sinks: Unconditional Linear Bounds and Positional Geometry**
– D. Vasilenko, A. Grabovoy. Submitted Sep. 2026, under review at AISTATS 2027.
– Removed the zero-BOS-value assumption: $\Omega(T)$ MLP cost for every value-output map via a geometric lemma on RMS-normalized vectors with a common shift; complete sink cost $\Theta(\kappa T / \sqrt{d} + T)$; stability under near-orthogonal positions and a rank-two construction showing the linear order is not universal.
- Bensemble: A Modular Python Library for Bayesian Deep Learning** | JOSS, in review | GitHub Jul. 2026
– D. Vasilenko, M. Nabiev, F. Sobolevsky, V. Kasiuk, O. Bakhteev.
– Open-source PyTorch library with a unified interface for 8 Bayesian deep learning and ensembling methods (Deep Ensembles, MC Dropout, VI, Laplace, PBP, NES); validated on CIFAR-10 vs. SVHN OOD detection.

RESEARCH EXPERIENCE

- Yandex R&D, YandexGPT Pretraining Team** Moscow, Russia
Research Intern Aug. 2026 – Present
– Mid-training data research (factual knowledge and agentic trajectories) for the AliceAI / YandexGPT foundation models within the Knowledge Research group.
- Intelligent Systems Department, MIPT** Moscow, Russia
Graduate Researcher Sep. 2025 – Present
– Researching theoretical stability and dynamics of Recurrent Transformers under the supervision of Dr. Andrey Grabovoy.
– Investigating Jacobian dynamics and spectral properties of deep Transformer components (MSA, FFN, RMSNorm).
- Moscow Institute of Physics and Technology (MIPT)** Moscow, Russia
Undergraduate Researcher Apr. 2024 – July 2025
– Bachelor's thesis: C++ numerical solvers for hyperbolic PDEs (method of characteristics).

TECHNICAL SKILLS

- Programming:** Python, C++
ML Frameworks: PyTorch, NumPy
Developer Tools: Git, Linux, Docker, LaTeX
Natural Languages: Russian (Native), English (Advanced / C1), German (A2)